

Yurt Portal: A Nomadic Spatial Interface to Reflect Cross-Cultural Virtual Digital Twins

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Figure 1: Overview of the Yurt Portal experience: (left) human mirroring mode from Japanese to Kyrgyz nomad cultural twin, (middle) object mirroring mode from Kyrgyz object to Japanese object, (right) Yurt Portal with virtual portal inside

Abstract

Yurt Portal is an interactive Extended Reality (XR) experience that augments physical space by generating and mirroring cultural digital twins to foster cross-cultural communication. Set in a yurt - a traditional Kyrgyz nomadic dwelling - the installation integrates physical objects from both Kyrgyz and Japanese cultures. Users engage with generative AI-powered digital twins of themselves and the objects, creating layered interactions between physical and virtual elements. The system comprises three key components: the yurt space, curated cultural objects, and AI-driven digital interactive

augmentations. By situating virtual content within the physical environment, the project explores the reality–virtuality continuum, positioning XR not merely as a technological medium but as a cultural interface. Yurt Portal emphasizes empathy-based cultural learning [Geertz 2017] through familiar artifacts, encouraging deeper understanding between underrepresented and well-known cultures. It embedded cultural narratives into the continuum of XR, highlighting how familiarity can foster empathy and build respect for cultural differences.

CCS Concepts

• **Human-centered computing** → **Interaction design**; **Mixed / augmented reality**.

Keywords

Cultural Heritage, Interactive Space, Extended Reality

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1 Introduction

Cultural traditions around the world offer unique perspectives, fostering discovery and cross-cultural understanding. Traditionally, culture sharing occurs through oral, visual, and written storytelling. These traditions have since been extended through the development of museums, exhibitions, and, more recently, the incorporation of AI-driven interactive technologies such as XR [Lindley and Whitham 2025; Wampfler et al. 2024] with interactive augmented environments [Gyls 2021], and generative design tools [Hung et al. 2025]. However, most cross-cultural exploration focuses on analyzing differences rather than drawing connections to one's own cultural background. To overcome this, we created a Yurt Portal interactive experience that emphasizes cultural similarity by reflecting users' own traditions onto those of another culture [Kanimetova et al. 2025]. Since the Yurt Portal was exhibited in Japan, a cultural mirroring occurred between Kyrgyzstan and Japan. Although both countries are in Asia, Kyrgyz culture is shaped by nomadic traditions, while Japan's history has been defined by a settled, agriculture-based way of life, with its cultural and social structures developing around permanent settlements.

2 Yurt Portal System Design

The portal design for Kyrgyz nomadic culture involves creating a portal-like experience, akin to a nomadic, portable house or yurt, allowing participants to transform themselves into Kyrgyz nomads. The Yurt Portal has two features: *human mirroring* and *object mirroring*, as shown in the Figure 1. The system design is based on generative cultural style-transfer software and sensing mechanism that captures the user and creates a cultural twin in real time (Figure 2). We implemented a model with additional training sets specific to Kyrgyz culture, enabling the reflection of objects and the user's appearance in Kyrgyz style.

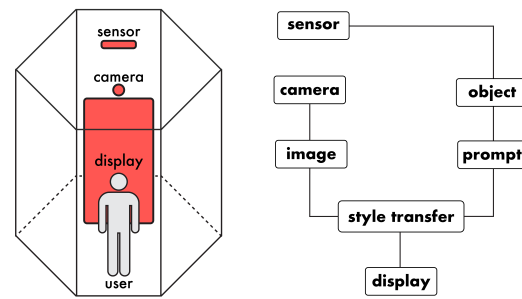


Figure 2: Yurt Portal System Configuration

3 User Experience

The experience begins with the user entering the yurt and choosing one of the two interactions: human or object mirroring. The interaction starts when the visitor picks up one of the physical objects presented inside the yurt. By showing cultural items to the camera, the user is able to see a mirrored version of the object and themselves in different cultural scenarios, celebrations, and experiences. For example, users can see themselves creating pottery art, enjoying Kyrgyz food, or participating in a traditional nomadic horse-riding game.

4 Conclusion

Yurt Portal reimagines XR as a medium for cultural empathy by immersing users within a mirrored space that combines familiar and unfamiliar traditions. By situating generative AI within a culturally curated XR environment, Yurt Portal transforms storytelling into an embodied dialogue, bridging cultural gaps and encouraging users to see themselves in others. Cross-cultural communication based on XR and generative AI, exemplified through Kyrgyz culture, opens up new possibilities for fostering cultural exchange worldwide.

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